

- Q.26 Write any five advantages and disadvantages of centrifugal separator.
- Q.27 Explain about secondary treatment of waste water in brief.
- Q.28 Discuss the method of desulphurization of fuel gas in brief.
- Q.29 Explain 'Composting' of solid waste.
- Q.30 Discuss the types of water pollution in detail.
- Q.31 Define particulate matter. Explain any four particulate matters.
- Q.32 Discuss 'Substitution of raw materials' as a source correction method.
- Q.33 Explain the control of air pollution by spray towers with diagram.
- Q.34 Differentiate between BOD and COD.
- Q.35 Explain flotation and sedimentation in detail.

#### SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What is desulphurization? Explain desulphurization of fuel oil in detail.
- Q.37 Discuss the sources of water pollution. Explain Trickling filter with neat & clean diagram.
- Q.38 Write the classification of solid waste. Explain incineration and land-filling in details.

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Roll No. ....

**6th Sem / Chem, P&P, Found. & Forg.**  
**Subject:- Pollution Control in Chemical Process**  
**Industry/ Env. Engg. & Safety**

Time : 3Hrs.

M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Auto mobile exhaust is example of \_\_\_\_\_ pollution.  
 a) Man-made                      b) Natural  
 c) thermal                        d) Nuclear
- Q.2 B.O.D is a term used to measure \_\_\_\_\_  
 a) Noise pollution              b) Water pollution  
 c) Air pollution                 d) Thermal pollution
- Q.3 \_\_\_\_\_ gas is most responsible to increase the average temperature of earth-  
 a) SO<sub>2</sub>                                b) SO<sub>3</sub>  
 c) CO<sub>2</sub>                               d) NO<sub>2</sub>
- Q.4 \_\_\_\_\_ can be produced from Land fill's waste  
 a) Biogas                            b) L.P.G gas  
 c) Natural gas                    d) All of these
- Q.5 Which of following is mainly responsible for causes of water pollution-?  
 a) Oil refineries                  b) afforestation  
 c) Textile mills                    d) Both (a) & (c)

- Q.6 Aerosol is a colloidal mixture of fine solid particles and \_\_\_\_\_.  
 a) Smoke  
 b) Nitrogen present in air  
 c) Liquid Droplets  
 d) All of these
- Q.7 Which source correction method is chosen when a particular raw material is responsible for causing pollution?  
 a) Raw material change  
 b) Equipment change  
 c) Process change  
 d) Emission treatment
- Q.8 Identify the wastes that are known as the municipal solid waste (MSW) from the following.  
 a) Wood pieces                      b) Plastic cans  
 c) Food wastes                      d) All of the above
- Q.9 Increases level of air pollution result in \_\_\_\_\_.  
 a) Global warming                  b) Soil erosion  
 c) Respiratory problems          d) All of the above
- Q.10 Which of the following air pollution control device has maximum efficiency?  
 a) Electrostatic precipitator  
 b) Spray tower  
 c) Dynamic precipitator  
 d) Cyclone scrubber

## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name any one method used for analysis of SO<sub>2</sub>.  
 Q.12 Write the full form of 'TDS'.  
 Q.13 Define P<sup>H</sup> & Write its range.  
 Q.14 Expand the term 'PAN'.  
 Q.15 Write the unit in which water hardness is expressed.  
 Q.16 Define Fog.  
 Q.17 Name any one method used for the collection of particulate pollutants.  
 Q.18 Name any one method used for source correction in air pollution control.  
 Q.19 Define industrial solid waste.  
 Q.20 Name any one method of disposal of solid waste.

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Discuss about grab sampling and stack sampling in brief.  
 Q.22 Discuss any five health hazards of solid waste pollution.  
 Q.23 Explain the method used for analysis of carbon-mono-oxide with diagram.  
 Q.24 Classify air pollution on the basis of origin and state of matter.  
 Q.25 Discuss the utility and significance of adsorption technique.